

Speaker Tracking: Segment-online Multi-talker Organization with a Varying Number of Speakers

Vahid A. Kalkhorani¹, Daniel Wong², Jacob Donley², Ashutosh Pandey², Buye Xu², and DeLiang Wang²

Abstract—Speaker tracking is the task of separating and following multiple speakers over time. It must address overlapped speech, speech onset and offset, talker identity, and time-varying speaker count. We propose a segment-online, modular framework for single- and multi-channel speaker tracking. The proposed system first performs speaker separation in each segment and computes speech activity via voice activity detection (VAD). To generate speaker tracks over time, we introduce a two-stage sequential organization strategy: Overlap-based stitching for continuous grouping and memory-based speaker verification for discontinuous grouping. For speaker separation, we employ complex spectral mapping to estimate the real and imaginary spectrograms of underlying speakers. The proposed system achieves state-of-the-art segment-online tracking performance on the LibriCSS and AMI datasets. Our framework significantly reduces diarization error rate (DER) and concatenated minimum-permutation word error rate (cpWER) compared to other methods.

Index Terms—Speaker tracking, speaker separation, speaker diarization, computational auditory scene analysis, complex spectral mapping.

I. INTRODUCTION

Speaker tracking is the broad task of separating overlapped talkers and following them over an extended period. Inspired by auditory scene analysis principles [1], we formulate speaker tracking as two related processes: simultaneous grouping or organization, and sequential organization (see also [2]). The simultaneous process deals with an acoustic mixture over a short time, or a segment of typically a few seconds long. When multiple speakers are active concurrently, simultaneous organization aims to separate this mixture into simultaneous streams, each corresponding to one speaker in the segment. On the other hand, sequential organization integrates simultaneous streams across segments to form speaker tracks, each corresponding to a distinct speaker over a longer period of time (such as a meeting). Sequential organization provides the mechanism of grouping the discontinuous utterances of the same talker.

Human listeners are remarkably adept at tracking speakers in noisy and dynamic environments [3], which is essentially the cocktail party problem [4]. In contrast, computational speaker tracking remains a formidable challenge [5, 6]. Real-world acoustic scenes include background noise and room

reverberation, which degrade signal quality and obscure speech cues [7]. Overlapped speech from concurrent talkers further complicates the task by blending the talker characteristics of overlapped speakers. In addition, speaker tracking has to maintain talker identity across discontinuous time periods, hence requiring a form of memory for extended durations and speaker modeling, and detecting new speakers.

Speaker tracking is related to, but broader than, speaker separation and speaker diarization. Speaker separation focuses on isolating concurrent voices within a specific segment, corresponding to simultaneous organization [7]. Speaker diarization is capable of linking same-talker utterances across time [8, 9]. A recent diarization study also uses the term speaker tracking to denote following many speakers over time [10]. However, diarization systems usually do not deal with overlapped speech, or label it as belonging to more than one talker without separating the underlying signals. Speaker tracking inherently addresses both tasks, separating overlapped speech and diarizing separated speech. The objective of speaker tracking is to generate the separated and enhanced speech signal of each speaker throughout their presence in an acoustic scene.

Deep-learning-based speaker separation has witnessed major advances in recent years [7]. Models such as deep clustering [11] and permutation invariant training (PIT) [12] have successfully addressed the permutation ambiguity issue in talker-independent separation in monaural settings. Subsequently, time-domain methods, including Conv-TasNet [13], were introduced to replace the short-time Fourier transform (STFT) with learnable encoders. A recent trend is the resurgence of time-frequency (T-F) domain approaches based on complex spectral mapping (CSM), which trains a deep neural network (DNN) to predict real and imaginary spectrograms [14]. CSM can fundamentally represent both signal magnitude and phase, and hence naturally encodes inter-channel phase relations of sound sources in multi-channel recordings, providing an important spatial dimension. CSM models, such as TF-GridNet [15], SpatialNet [16], and TF-CrossNet [17], have achieved state-of-the-art speaker separation performance.

To move toward real-world applicability, subsequent work has proposed continuous speaker separation (CSS) to separate speaker streams in a conversational setting [18–23]. CSS processes a sliding sequence of partially overlapped segments and outputs a fixed number of overlap-free streams, irrespective of the total speaker count. CSS addresses simultaneous grouping within each segment and stitches the same-talker signal across consecutive segments, but does not perform sequential grouping across discontinuous segments. Consequently, a single output stream may contain speech from multiple speakers at different

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times. Such sequential grouping requires a separate diarization backend. Previous systems have combined CSS and speaker diarization [24–27]. Yoshioka et al. [24] integrate CSS with online audio-visual diarization in a separate-recognize-diarize pipeline, but rely on a camera feed and a prior list of meeting attendees. Von Neumann et al. [26] combine CSS with offline d-vector clustering and ASR-supported speaker turn detection, which achieves strong concatenated minimum-permutation word error rate (cpWER) on LibriCSS, but operates offline.

Traditional speaker diarization pipelines use voice activity detection (VAD), embedding vector extraction, and clustering [28–30]. While effective offline, these systems often assume a fixed speaker count and struggle with speaker overlaps. Overlap-aware methods, including end-to-end neural diarization (EEND) models, jointly perform detection and attribution [31–33], with extensions to unknown speaker counts [34–36]. Nevertheless, diarization models do not separate overlapped speech signals, limiting their utility for downstream tasks such as ASR.

An alternative to CSS, termed speaker separation via neural diarization (SSND) [37], employs an offline EEND to estimate speech activities and condition a downstream speaker separation model. Assuming a maximum of two overlapped talkers at each time-frame, the speaker separation model in this framework outputs two continuous streams regardless of the total speaker count. SSND extracts speaker embeddings from non-overlapped segments identified by the diarization module. These embeddings, combined with the diarization results, guide the assignment of separated signals to the output streams as a post-processing step. DCF-DS [38] further integrates diarization and separation through a deep cascade fusion mechanism, allowing each module to condition on the other, and was recently extended to exploit multi-channel spatial information [39]. However, like SSND, DCF-DS is an offline model and needs prior speaker diarization results.

Despite these advances, a significant research gap remains between the current separation and diarization results and online speaker tracking. Existing systems often rely on offline backends for sequential organization or require prior speaker enrollment to guide separation. Little prior research aims to provide a unified system for single-channel and multi-channel segment-online speaker tracking without assuming a fixed number of speakers. Addressing these limitations would require a broad framework that deals with both simultaneous and sequential organization. Sequential organization would need to be mediated by a form of working memory, referring to temporary storage of information for performing complex tasks [40].

This paper tackles the speaker tracking challenge by developing a segment-online, modular framework capable of both simultaneous and sequential organization with an unknown number of speakers. Simultaneous organization within a segment will be conducted through speaker separation based on complex spectral mapping, producing simultaneous streams. We propose to address sequential organization across different segments in two stages. The first stage performs overlap-based stitching to organize simultaneous streams, to extend simultaneous streams across consecutive segments. The second stage sequentially organizes discontinuous simultaneous

streams into speaker tracks by introducing a working memory mechanism based on speaker verification. The proposed two-stage approach addresses the limitations of cascade systems: Stitching based on spectral overlap maintains the temporal continuity of speech from one segment to the next, and memory-driven verification maintains talker identity over extended periods of no speech activity. Unlike the SSND and DCF-DS models, our framework groups the discontinuous speech signals of the same talker without a separate offline backend or prior speaker enrollment. We present both single-channel and multi-channel tracking algorithms. In single-channel scenarios, tracking relies on spectral, temporal, and talker cues. In the multi-channel setup, tracking further exploits spatial cues to improve both simultaneous and sequential organization. We systematically evaluate the proposed framework on the LibriCSS [18] and AMI [41] datasets.

The primary contributions of this paper are summarized as follows. First, we propose a segment-online speaker tracking framework that integrates speaker separation and sequential organization in a modular fashion. Second, we propose a two-stage sequential organization approach, where a working memory mechanism is utilized to organize temporally-discontinuous utterances into distinct speakers. Third, the proposed single-channel and multi-channel speaker tracking algorithms advance the state-of-the-art performance on the LibriCSS and AMI datasets.

The rest of the paper is organized as follows. Section II describes the proposed speaker tracking framework. Section III presents the experimental setup. Section IV describes the evaluation and comparison results. Section V concludes the paper.

II. MODEL DESCRIPTION

A varying number of active speakers poses a considerable challenge to speech separation, as DNN models would require a fixed output dimension. In real-world environments, speakers may enter or exit an auditory scene at arbitrary times, resulting in varying numbers of concurrent sources and disjoint speaker tracks. A robust tracking system must therefore function without prior knowledge of the total number of speakers in the scene, and maintain consistent speaker identities across the entire session.

To achieve this, the proposed tracking system, denoted by $\mathcal{F}_{\text{track}}$, maps the input observations $\mathbf{y}(t) \in \mathbb{R}^M$ in a session to a set of estimated speaker streams $\{\hat{\mathbf{s}}_n(t)\}_{n=0}^{N(t)}$, where M is the number of microphones, t is the discrete time index, n is the globally consistent speaker index, and $N(t)$ is the total number of distinct speakers tracked up to time t

$$\{\hat{\mathbf{s}}_n(t)\}_{n=0}^{N(t)} = \mathcal{F}_{\text{track}}(\mathbf{y}(t)). \quad (1)$$

In this formulation, $N(t) = 0$ indicates that no active speaker has been tracked by time t , e.g., during the session beginning, and $\hat{\mathbf{s}}_0(t)$ denotes the nonspeech background signal.

When $M = 1$ (single-channel), the system must rely on spectro-temporal patterns to distinguish speakers. In the multi-channel case, $M > 1$, the system additionally exploits spatial cues, such as inter-channel phase difference, to spatially filter

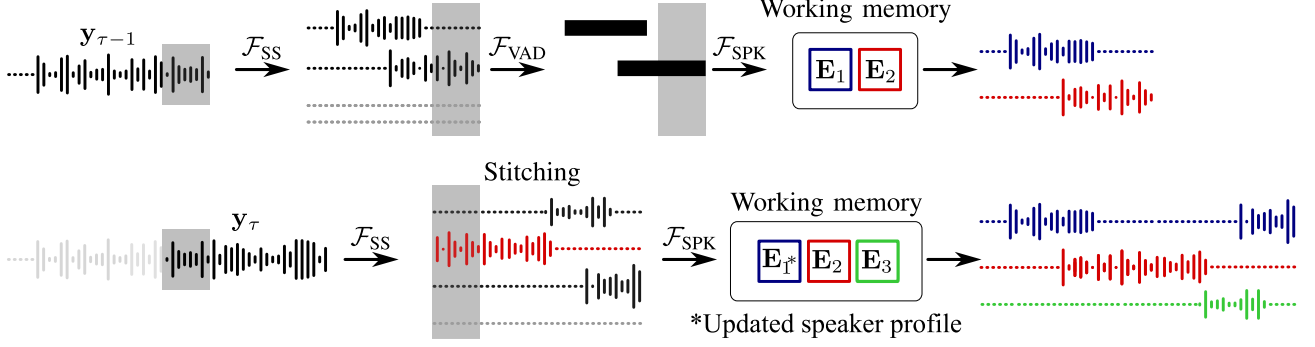


Fig. 1: Illustration of the proposed modular speaker tracking framework. The architecture processes the input as sliding segments in different stages: speaker separation for simultaneous organization within each segment, VAD to compute activity traces, overlap-based stitching to group simultaneous streams across consecutive segments, and a working memory mechanism based on speaker embeddings to sequentially organize discontinuous streams into speaker tracks. Black waveforms denote the mixture and separated speakers within individual segments before sequential organization. A color denotes a distinct speaker track after sequential organization. Gray regions mark overlap intervals.

and track sources. The system must maintain global speaker correspondence to ensure that the estimate $\hat{s}_n(t)$ assigned to speaker n corresponds to the same talker at all times, regardless of overlaps, interruptions, or silences. Sequential organization is required to prevent talker label switching across segments, particularly when multiple speakers are active (i.e., talking) simultaneously.

A. Speaker separation for simultaneous grouping

We propose a modular segment-online architecture starting with speaker separation, as shown in Fig. 1. The input audio is divided into overlapped segments indexed by τ . A separation model $\mathcal{F}_{SS}$ estimates a fixed number of candidate streams

$$\{\hat{\mathbf{s}}_{\tau,k}\}_{k=1}^K = \mathcal{F}_{SS}(\mathbf{y}_{\tau}), \quad (2)$$

where k indexes speakers within segment τ , K is a hyperparameter denoting the maximum number of separated speakers the speaker separation model can represent in a segment, $\mathbf{y}_{\tau} \in \mathbb{R}^{M \times L}$ is the mixture in segment τ , L denotes the number of samples in a segment, and $\hat{\mathbf{s}}_{\tau,k} \in \mathbb{R}^L$ denotes the estimated signal for speaker k in segment τ . When a segment contains fewer than K talkers, some of the candidate streams will be a zero signal.

While the speaker separation model can effectively separate simultaneous talkers within each segment, it does not maintain speaker identity across segments. Therefore, additional processing is required to associate separated streams over time to ensure consistent speaker labeling. This association relies on both continuous and discontinuous organization, as detailed in Sec. II-C and Sec. II-D.

To resolve permutation ambiguity during the training, we utilize PIT [12]. Since some speakers may be silent, the training objective should take into account inactive candidate streams. We therefore propose scale-invariant source-aggregated signal-to-distortion ratio (SI-SA-SDR) as the primary loss function. This formulation extends source-aggregated signal-to-distortion ratio (SA-SDR) [42] by introducing scale-invariance as follows:

$$\mathcal{L}_{\text{SI-SA-SDR}} = -10 \log_{10} \frac{\sum_{k=1}^K \|\alpha \mathbf{s}_{\tau,k}\|_2^2}{\sum_{k=1}^K \|\alpha \mathbf{s}_{\tau,k} - \hat{\mathbf{s}}_{\tau,k}\|_2^2}, \quad (3)$$

where $\mathbf{s}_{\tau,k}$ is the clean (reference) signal for speaker k in segment τ , $\|\cdot\|_2$ is the L_2 norm operator, and α is a single scalar value shared across all K speakers, calculated as

$$\alpha = \frac{\sum_{k=1}^K \hat{\mathbf{s}}_{\tau,k}^T \mathbf{s}_{\tau,k}}{\sum_{k=1}^K \hat{\mathbf{s}}_{\tau,k}^T \hat{\mathbf{s}}_{\tau,k}}. \quad (4)$$

To further suppress residual energy in inactive outputs, we include a magnitude reconstruction loss. To ensure consistency between the complex domain predictions and the resulting waveforms, we compute the magnitude loss of the reconstructed signals:

$$\mathcal{L}_{\text{Mag}} = \sum_{k=1}^K \left\| |\mathbf{s}_{\tau,k}| - |\hat{\mathbf{s}}_{\tau,k}| \right\|_1, \quad (5)$$

where $|\mathbf{s}_{\tau,k}|$ and $|\hat{\mathbf{s}}_{\tau,k}|$ denote the magnitude of reference and estimated spectrograms in segment τ of speaker k , respectively, and $\|\cdot\|_1$ is the L_1 norm operator.

The final loss is the sum of the two terms:

$$\mathcal{L} = \mathcal{L}_{\text{SI-SA-SDR}} + \mathcal{L}_{\text{Mag}}. \quad (6)$$

This combined objective yields stable training and strong separation quality and downstream automatic speech recognition (ASR) performance under reverberant, multi-talker conditions.

B. Voice activity detection

Following separation, we compute frame-level activity traces for each separated source using a pre-trained VAD model, denoted as $\mathcal{F}_{\text{VAD}}$ in Fig. 1. These activity labels are used to remove speaker separation leakage errors, determine which speakers are active in the overlap interval between consecutive segments, and to restrict comparisons during overlap-based stitching. For the final diarization output, the VAD traces of each speaker from all segments are concatenated.

C. Sequential organization in consecutive segments via stitching

The separation stage yields a set of local, segment-level speaker estimates. To form coherent speaker tracks, these local estimates must be sequentially organized. We propose a two-stage sequential organization strategy that addresses both short- and long-term sequential organization.

For consecutive segments $\mathbf{y}_{\tau-1}$ and $\mathbf{y}_\tau$, we utilize the overlapped interval of length L_{ovl} samples to enforce temporal continuity. Since the acoustic realization of a separated speaker should remain stable over this short interval, we perform sequential grouping by matching the separated waveforms directly; in other words, the overlapped interval provides the “thread” for a stitching process.

Specifically, let $\hat{\mathbf{s}}_{\tau,k} \in \mathbb{R}^L$ denote the estimated signal for speaker $k \in \{1, \dots, K\}$ in segment τ . We denote the first L_{ovl} samples of $\hat{\mathbf{s}}_{\tau,k}$ as $\mathbf{p}_{\tau,k} \in \mathbb{R}^{L_{\text{ovl}}}$ and the last L_{ovl} samples as $\mathbf{q}_{\tau,k} \in \mathbb{R}^{L_{\text{ovl}}}$. Let $\mathcal{A}_{\tau-1}, \mathcal{A}_\tau \subseteq \{1, \dots, K\}$ denote the sets of active speakers separated within the overlapped interval by $\mathcal{F}_{\text{SS}}$ in segments $\tau-1$ and τ , respectively. For each speaker pair $(k', k) \in \mathcal{A}_{\tau-1} \times \mathcal{A}_\tau$, we first compute the pairwise distance

$$d_{k',k} = \mathcal{D}(\mathbf{q}_{\tau-1,k'}, \mathbf{p}_{\tau,k}), \quad (7)$$

where $\mathcal{D}$ is a distance metric. If $d_{k',k} > \gamma$, where γ denotes the threshold of acceptable dissimilarity, the pair (k', k) is removed. Afterwards, we remove speaker k from $\mathcal{A}_\tau$ if all the pairs involving k are removed, and speaker k' likewise.

Next, we construct a set of all possible stitches $\mathcal{P}$ over the remaining active speakers in $\mathcal{A}_\tau$ and $\mathcal{A}_{\tau-1}$, where each $\pi \in \mathcal{P}$ is a stitch forming a specific pairing or permutation between all the remaining speakers in $\mathcal{A}_\tau$ and the remaining ones in $\mathcal{A}_{\tau-1}$. The optimal stitch π^* is selected as

$$\pi^* = \arg \min_{\pi \in \mathcal{P}} \sum_{(k',k) \in \pi} d_{k',k}. \quad (8)$$

For each pair $(k', k) \in \pi^*$, the separated stream of speaker k in segment τ is stitched to that of speaker k' in segment $\tau-1$. Specifically, $\mathbf{q}_{\tau-1,k'}$ and $\mathbf{p}_{\tau,k}$ are combined via an overlap-and-add technique. To smooth segment boundaries, we first apply complementary linear ramps that fade out $\mathbf{q}_{\tau-1,k'}$ and fade in $\mathbf{p}_{\tau,k}$, and then sum the two ramped signals to form the new speaker signal in the overlapped interval. The remaining non-overlapped portion of $\hat{\mathbf{s}}_{\tau,k}$ is appended to produce the stitched stream. All active streams in segment τ that are not stitched to those in $\tau-1$ will be further organized as described in Sec. II-D.

Although the idea of stitching is intuitive and has been used previously [43, 18], we cannot find a quantitative description and there are various ways of implementing the stitching idea. The stitching process described above is accurate and computationally efficient for continuous grouping. But it fails when speakers pause longer than L_{ovl} or are not active in the overlapped interval, necessitating discontinuous grouping.

D. Sequential organization in disjoint segments via working memory

In human cognition [40] working memory (WM) serves to temporarily hold and manipulate information necessary for

complex reasoning and tracking. Motivated by this form of human memory, our framework incorporates a WM mechanism to perform discontinuous organization. The primary role of this mechanism is to maintain speaker identity beyond consecutive segments. While stitching effectively handles sequential grouping in consecutive segments, it fails when a speaker pauses for longer than a segment or is inactive in overlapped intervals. To perform sequential grouping in disjoint segments, the WM mechanism includes dynamically formed speaker models $\mathcal{F}_{\text{SPK}}$. When a speaker stream in the current segment τ cannot be grouped to earlier speakers via stitching, we extract a speaker embedding vector

$$\mathbf{e}_{\tau,k} = \mathcal{F}_{\text{SPK}}(\hat{\mathbf{s}}_{\tau,k}). \quad (9)$$

The model $\mathcal{F}_{\text{SPK}}$ is a pre-trained speaker encoder (e.g., ReD-ImNet [44]) that maps variable-length audio segments to a fixed-dimensional discriminative embedding space.

The proposed working memory mechanism maintains a set of global speaker profiles $\mathcal{E}(t) = \{\mathbf{E}_n\}_{n=1}^{N(t)}$. The newly extracted embedding $\mathbf{e}_{\tau,k}$ is compared against all the profiles in the WM using the cosine similarity score:

$$\mathcal{C}(\mathbf{e}_{\tau,k}, \mathbf{E}_n) = \frac{\mathbf{e}_{\tau,k}^\top \mathbf{E}_n}{\|\mathbf{e}_{\tau,k}\|_2 \|\mathbf{E}_n\|_2}. \quad (10)$$

For each unstitched speaker in segment τ , we identify the most similar global speaker profile as

$$n^* = \arg \max_{n \in \{1, \dots, N(t)\}} \mathcal{C}(\mathbf{e}_{\tau,k}, \mathbf{E}_n). \quad (11)$$

If $\mathcal{C}(\mathbf{e}_{\tau,k}, \mathbf{E}_{n^*}) \geq \theta$, the stream is assigned to speaker n^* and the profile $\mathbf{E}_{n^*}$ is updated using the moving average method (denoted as $\mathbf{E}_{1*}$ in Fig. 1); otherwise, a new speaker profile is initialized and added to the WM (e.g., $\mathbf{E}_3$ in Fig. 1). By dynamically updating these profiles, the WM acts as an adaptable speaker buffer. This WM mechanism enables the system to maintain speaker identities across interruptions and silences without prior knowledge of the total number of speakers, bridging the gap between simultaneously organized streams and coherent speaker tracks.

Existing buffer-based methods commonly store mixture signals or embeddings for repeated evaluations. Examples include the speaker-tracing buffer in SA-EEND [45], block-wise EEND [46], and the arrival-order speaker cache in streaming Sortformer [47]. These methods concatenate incoming features with buffered representations and reprocess the entire sequence for each new chunk. This repeated processing increases computational cost and introduces variable-length inputs that require variable chunk-size training [46]. In addition, feature buffers may contain overlapped speech depending on the frame selection policy, and the total number of output speakers may be restricted. In contrast, our WM mechanism maintains compact, speaker-specific embedding profiles extracted from separated speech. Our framework only limits the maximum number of simultaneously active speakers (i.e., K), and allows the total number of global speaker tracks to grow dynamically. Leveraging continuous grouping via stitching and discontinuous grouping via speaker profiling, our approach completely avoids repeated inference over past segments.

E. Implementation details

We adopt TF-CrossNet [17] as the speaker separation model. TF-CrossNet performs CSM in the T-F domain and predicts the real and imaginary parts of each speaker’s spectrogram. The TF-CrossNet architecture comprises five modules: (i) an encoder layer to project mixture real and imaginary (RI) representations; (ii) a global multi-head self-attention (GMHSA) module to model cross-frame and cross-band dependencies; (iii) a cross-band module to capture inter-frequency correlations across all channels; (iv) a narrow-band module to model fine-grained structure within adjacent frequency bins; and (v) a decoder that yields separated spectrograms, which are converted back to time-domain waveforms via inverse STFT [17]. TF-CrossNet also incorporates random-chunk positional encoding (RCPE) to mitigate performance degradation on long utterances and improve temporal generalization [17]; however, we do not utilize RCPE here as we operate on relatively short segments. See [17] for further details. For evaluations on the AMI dataset, the speaker separation model is further fine-tuned on the AMI training set for 10 epochs following training on a simulated dataset (see Sec. III-A).

We utilize three pre-trained VAD models: Pyannote [48], Silero VAD¹, and WebRTC². Pyannote employs a neural sequence labeling architecture trained on a diverse set of conversational recordings and provides frame-level speech activity scores. Silero is a lightweight, real-time model optimized for low-latency applications, producing binary decisions at high temporal resolution. WebRTC is a signal processing based model that classifies short frames as speech or non-speech using a Gaussian mixture model, and is computationally efficient.

For the stitching mechanism for continuous grouping, we use the negative scale-invariant signal-to-distortion ratio (SI-SDR) as the distance measure $\mathcal{D}$ in Eq. 7, and set the threshold γ to -5 dB.

We experiment with five speaker verification models: NeMo [49], ReDimNet [44], ReDimNet2 [50], SpeechBrain [51], and Wespeaker [52]. Each model extracts a fixed-dimensional embedding from a speech segment, which is then compared to previously-tracked speakers the using cosine similarity in Eq. 10. If the maximal similarity score exceeds a predefined threshold, the speech segment is added to the track of the closest talker; otherwise, a new track is initiated. To determine the optimal threshold θ for each model, we evaluate speaker verification accuracy on 3000 speaker pairs randomly sampled from LibriSpeech development subset [53], with positive (same-speaker) and negative (different-speaker) pairs sampled with equal probability. We then perform a grid search over θ ranging from 0.05 to 0.95 in steps of 0.05, and select the threshold that yields the highest verification accuracy. For ReDimNet, we utilize the pre-trained M-sized model, which is trained on the combined VoxBlink2 [54], VoxCeleb2 [55], and CN-Celeb [56] datasets³, and for ReDimNet2 we select the pre-trained B6 model which is trained on VoxCeleb2 dataset⁴.

¹<https://github.com/snakers4/silero-vad>

²<https://github.com/wiseman/py-webrtcvad/>

³<https://github.com/IDRnD/ReDimNet/>

⁴<https://github.com/PalabraAI/redimnet2/>

In all cases, the speaker model weights are frozen and we do not fine-tune them.

III. EXPERIMENTAL SETUP

A. Datasets

To train the proposed speaker tracking system, we generate simulated mixtures that reflect the acoustic characteristics of two evaluation datasets: the LibriCSS [18] and the AMI [41] meeting datasets. For both configurations, each mixture includes at most four speakers drawn from the *train-clean-100* and *train-clean-360* subsets of LibriSpeech [53]. Room impulse responses (RIRs) are synthesized using the *gpuRIR* [57] toolkit, with 100,000 and 2,500 rooms simulated for the training and validation sets, respectively.

For the LibriCSS simulation, room dimensions are sampled uniformly with lengths and widths ranging from 3 to 8 m and heights from 3 to 4 m. The reverberation time (RT_{60}) is drawn from a truncated Gaussian distribution ($\mu = 0.4$ s, $\sigma = 0.12$ s) bounded between 0.1 and 0.6 s. We construct a 7-channel microphone array corresponding to the LibriCSS hardware geometry (radius 4.25 cm), positioned within 1.0 m of the room center at an elevation between 1.0 and 1.5 m. Speakers are distributed at radial distances of 0.5 to 4.5 m from the array, with heights ranging from 1.0 to 1.5 m. To ensure sufficient spatial separability, a minimum azimuth difference of 5° is maintained across all sources.

Following the LibriCSS recipe [18]⁵, we simulate meeting-style conversations where two to four speakers each provide two utterances from the LibriSpeech training set [53]. Each conversation is simulated as follows. First, an utterance is randomly drawn from the set of the utterances from all speakers in a conversation. Then a silence of a random duration between 0.5 s to 3.0 s is added with a 26% probability. If a silence is added, the next utterance is drawn from the set of the remaining utterances. If no silence is added, the next utterance is drawn from the set of the remaining utterances with the condition that the drawn utterance cannot be from the same talker as the current utterance. Then the current and next utterance are mixed with a speaker overlap ratio randomly selected from 0% to 45%. This process continues until all the utterances of the set are used up. The simulated meetings are partitioned into 8 s segments for speaker separation model training.

We evaluate speaker tracking in both session- and section-wise scenarios. In session-wise evaluation, each ten-minute mini-session signal is processed directly. For section-wise evaluation, we utilize the pre-segmented LibriCSS sections. These sections range from 5 to 120 seconds in length and contain between 2 and 10 utterances from up to 8 distinct speakers.

For AMI data simulation, we generate RIRs modeling standard conference rooms. Room dimensions are sampled uniformly with lengths and widths spanning 3 to 10 m and heights from 3 to 4 m. RT_{60} is drawn from the same truncated Gaussian distribution but bounded between 0.2 and 0.7 s to reflect heavier reverberation. We simulate a single microphone

⁵Available at: https://github.com/jsalt2020-asrdiar/jsalt2020_simulate

positioned near the room center at a vertical height of 0.7 to 0.85 m to represent a table-mounted microphone. Up to four simultaneous speakers are distributed around the microphone with radii from 0.75 to 2.5 m and elevations from 1.1 to 1.3 m, representing seated participants. The 5° minimum azimuth separation is also enforced across all localized sources.

In both simulation protocols, diffuse background noise is generated using the method described in Habets *et al.* [58], with single-channel ambient noises drawn from the REVERB challenge dataset [59]. The mixture signal-to-noise ratio (SNR) is randomly selected between 5 and 20 dB. To support both single- and multi-channel configurations, we adapt array usage accordingly: for multi-channel models, the full microphone array is utilized, whereas for single-channel models, we randomly select a single channel for training and evaluate using the primary reference channel.

B. Evaluation metrics

We use diarization error rate (DER) to assess diarization performance, and speaker-attributed cpWER to assess ASR performance. DER quantifies diarization accuracy as the fraction of reference speech duration attributed to missed speech (MI), false alarm (FA), or speaker confusion (CF) errors:

$$DER = \frac{T_{MI} + T_{FA} + T_{CF}}{T_{\text{total}}}, \quad (12)$$

where T_{MI} , T_{FA} , and T_{CF} are the durations of missed speech, false alarm, and speaker confusion, respectively, and T_{total} is the total reference speech duration. A 0-s collar tolerance is applied in all evaluations.

Word error rate (WER) [18] measures transcription accuracy without enforcing speaker identity. Reference and hypothesis words are pooled across all speakers, and WER is defined as

$$WER = \frac{W^S + W^D + W^I}{W}, \quad (13)$$

where W^S , W^D , and W^I are the numbers of substitution, deletion, and insertion word errors, respectively, and W is the total number of reference words.

Metric cpWER [6] extends WER to the multi-talker setting by jointly optimizing all speaker assignments. The utterances of each talker are first concatenated into a single stream for both reference and hypothesis. WER is then evaluated over all possible speaker assignments $\pi \in \mathcal{Q}$ of the hypothesis, and the minimum is taken as the final score:

$$cpWER = \min_{\pi \in \mathcal{Q}} \frac{\sum_{(n,n') \in \pi} (W_{n,n'}^S + W_{n,n'}^D + W_{n,n'}^I)}{\sum_n W_n}, \quad (14)$$

where $\mathcal{Q}$ is the set of all pairings between reference and hypothesis speakers, $(n, n') \in \pi$ denotes a hypothesized pair of reference speaker n and hypothesis speaker n' , $W_{n,n'}^S$, $W_{n,n'}^D$, and $W_{n,n'}^I$ are the substitution, deletion, and insertion errors when reference speaker n is matched to hypothesis speaker n' , and W_n is the number of words spoken by reference speaker n .

To properly penalize speaker count mismatches during evaluation, we pad the smaller speaker set with empty speaker

streams prior to computing metrics. Specifically, we add zero-valued signals for DER and empty transcripts for cpWER. This ensures unmatched hypothesis speakers count as false alarms or insertions, and unmatched reference speakers count as missed speech or deletions.

IV. EVALUATION RESULTS AND COMPARISONS

A. Results on LibriCSS

We first evaluate our system on the LibriCSS dataset using DER, and cpWER as metrics. Table I reports DER results across all overlap conditions. Existing comparison approaches fall into two categories: (i) systems that apply a separation or beamforming front-end followed by diarization, and (ii) systems that directly perform diarization, i.e., without front-end separation.

The first category includes mask-based MVDR [18], MIMO-BF-MISO [61], and MISO-BF-MISO [19], which combine spatial filtering and clustering-based diarization (i.e., x-vector + spectral clustering). While effective in reducing overlap-caused errors, these systems are not designed for online operation. In addition, LlaSep [62] performs generative separation using an autoregressive audio language model and determines speaker activity from the separated streams via energy-based thresholding, achieving an average DER of 23.43%. The second category consists of diarization models without speaker separation, including x-vector + SC, AHC, VBx [25], and SC-EEND [37]. These models are simpler but tend to degrade more severely under high overlap conditions due to the absence of a separation stage. Competitive methods include RPN and TS-VAD [25], which leverage speaker-specific enrollment and target speaker detection networks. These methods achieve low DER (e.g., 7.6% for TS-VAD) but rely on oracle knowledge of the number of speakers, an assumption not made in our framework.

The proposed single-channel model achieves 4.63% DER. This represents an absolute reduction of more than 13.5% relative to the x-vector + SC baseline [25] (18.19%). The proposed seven-channel model further reduces the DER to the new state-of-the-art score of 3.38%, outperforming the previous best MC-EEND [37] (4.68%), despite MC-EEND operating offline.

To isolate the effects of speaker separation from subsequent sequential organization, we report results under an ‘‘Oracle’’ condition. In this setup, we utilize reference dry signals (*each_spk* files) from LibriCSS as separated speech for two-stage sequential grouping. The oracle results provide the lower bound on errors caused by VAD, speaker models, and the ASR backend. As shown in Table I, our 7-channel DER of 3.38% is very close to the oracle DER of 2.98%, demonstrating the high separation quality of TF-CrossNet.

For the ASR task, we report speaker-attributed cpWER in Table II. While standard LibriCSS evaluations often prioritize WER, cpWER provides a more meaningful assessment of speaker tracking by penalizing sequential grouping errors. Our proposed single-channel method achieves a cpWER of 9.25% with E2E ASR and 7.20% with E2E-SSL ASR. Our seven-channel method further reduces cpWER to 5.32% and 4.39%

TABLE I: DER results (in %) of the proposed and comparison diarization systems on LibriCSS session-wise evaluation.

Separation method	Diarization method	Overlap ratio						All
		OL	OS	10%	20%	30%	40%	
Mask-based MVDR [18]	X-vector+SC [25]	13.42	11.49	11.63	14.22	16.98	16.18	14.18
Multi-Frame MCWF [60]	X-vector+SC [25]	11.80	10.20	13.84	15.76	18.69	17.33	15.04
MISO-BF-MISO [19]	DOA-based [21]	10.69	11.95	11.25	12.22	13.04	14.31	12.36
MIMO-BF-MISO [61]	X-vector+SC [25]	10.34	10.4	8.97	9.52	11.66	9.54	9.9
-	MC-EEND [37]	6.12	4.94	3.36	4.09	4.88	5.05	4.68
-	X-vector+SC [25]	10.25	9.29	14.04	18.76	23.82	27.43	18.19
-	AHC [25]	16.1	12.0	16.9	23.6	28.3	33.2	22.6
-	VBx [25]	14.6	11.1	14.3	21.5	25.4	31.2	20.5
-	SC [25]	10.9	9.5	13.9	18.9	23.7	27.4	18.3
-	SC-EEND [37]	23.48	23.2	29.55	27.1	34.94	39.22	30.36
-	RPN*[25]	4.5	9.1	8.3	6.7	11.6	14.2	9.5
-	TS-VAD*[25]	6.0	4.6	6.6	7.3	10.3	9.5	7.6
LlaSep [62]	-	18.57	21.91	24.79	26.85	20.96	27.52	23.43
TF-CrossNet (1-ch)	Sequential grouping	3.43	3.57	4.63	4.20	6.05	5.25	4.63
TF-CrossNet (7-ch)	Sequential grouping	3.22	3.50	2.86	3.22	3.95	3.35	3.38
Oracle	Sequential grouping	3.12	2.83	2.69	2.63	3.68	2.74	2.98

* Assumes that the oracle number of speakers is known.

TABLE II: cpWER results (in %) for different separation and diarization methods on LibriCSS session-wise evaluation.

Separation method	Diarization method	ASR	DER (%)	Overlap ratio						All
				OL	OS	10%	20%	30%	40%	
Unprocessed	X-vector + SC	E2E	26.76	12.2	13.95	20.12	29.64	35.06	41.81	27.01
Mask-based MVDR [18]	X-vector + SC	E2E	14.18	13.07	8.78	10.51	15.37	17.54	17.63	14.13
MISO-BF-MISO [19]	X-vector + SC	E2E	8.33	9.24	10.84	11.05	12.27	16.39	17.42	12.87
MISO-BF-MISO [19]	DOA-based	E2E	12.36	9.03	10.18	10.89	13.6	16.14	18.05	12.98
MIMO-BF-MISO [61]	X-vector + SC	E2E	9.90	8.05	7.05	7.31	8.48	10.81	10.03	8.76
TF-GridNet [26]	ASR-based	E2E-SSL	9.4	-	-	-	-	-	-	7.2
DCF-DS [38]	Joint	E2E	-	-	-	-	-	-	-	6.30
SSND [37]	MC-EEND	E2E	7.62	3.52	5.56	3.98	4.76	5.58	6.55	5.13
SSND [37]	MC-EEND	E2E-SSL	7.62	1.97	3.67	2.46	3.06	3.52	4.07	3.22
TMS [63]	-	E2E-SSL	-	1.98	3.10	2.05	2.55	2.80	3.40	2.64
TF-CrossNet (1-ch) [17]	Sequential grouping	E2E	4.63	5.62	6.00	8.30	8.53	12.62	12.07	9.25
TF-CrossNet (1-ch) [17]	Sequential grouping	E2E-SSL	4.63	4.51	4.82	6.25	6.36	10.14	9.47	7.20
TF-CrossNet (7-ch) [17]	Sequential grouping	E2E	3.38	4.52	4.74	4.16	5.26	6.58	6.03	5.32
TF-CrossNet (7-ch) [17]	Sequential grouping	E2E-SSL	3.38	4.32	3.22	3.00	4.20	5.53	5.49	4.39
Oracle	Sequential grouping	E2E	2.98	3.39	2.75	3.10	3.01	5.78	3.53	3.66
Oracle	Sequential grouping	E2E-SSL	2.98	2.78	2.21	2.90	2.62	4.72	3.28	3.15

respectively, surpassing the MIMO-BF-MISO baseline (8.76 %) and achieving competitive performance to the offline SSND model (5.13% and 3.22%).

Compared to DCF-DS [38], our segment-online system avoids full-utterance context dependencies. DCF-DS requires a global diarization prior to segment the audio before window-based processing. Furthermore, its stitching relies on global speaker indices to reconstruct continuous streams from local separation results. Compared to TMS [63], our system generates multiple tracks simultaneously without requiring semantic cues. TMS runs an online ASR system on the raw mixture to generate preliminary transcripts as text prompts. It then isolates the acoustic signal that aligns with the prompt, treating other voices as background interference. This method involves a large language model and operates offline.

B. Results on AMI

We now evaluate our system on the AMI IHM-Mix meeting corpus using DER as the metric. Compared to LibriCSS, the

AMI corpus introduces severe challenges due to spontaneous conversational dynamics and real-room acoustics. AMI contains frequent short speech bursts that complicate speaker tracking. Furthermore, real-world reverberation and background noise degrade speaker separation quality. Additionally, AMI meetings average 33 minutes in duration, posing significant challenges for online diarization systems to maintain speaker identity over extended periods.

Table III reports the performances of the proposed and comparison systems. Among offline methods, AED-EEND-EE with Conformer [64] achieves the lowest DER of 13.00 %. Among online methods, our speaker tracking system obtains the best DER of 18.43%, outperforming LS-EEND [36] and other online alternatives. These results demonstrate the robustness of our speaker tracking framework in realistic meeting scenarios.

C. Ablation study

To isolate the effects of individual system components, we conduct an ablation study analyzing speaker models,

TABLE III: DER results (in %) of the proposed and comparison diarization systems on AMI evaluation set.

Methods	DER (%)
Offline	
X-vector AHC + VB + OVL [65, 66]	28.15
SA-EEND [32, 33]	27.70
Transcribe-to-Diarize [67]	24.43
Horiguchi <i>et al.</i> [68]	23.62
Multi-Class Spec-Clustering [69]	23.60
CmpEm + Overlap Detector [70]	22.93
EENDA-EDA [34, 33]	21.56
ANSD-MA-MSE [71]	16.95
AED-EEND-EE + Conformer [64]	13.00
Online	
Overlap-aware speaker embeddings [72]	30.4
SSep AMI + VAD E2E [73]	27.2
LS-EEND [36]	20.76
Speaker tracking	18.43

VAD models, segment sizes, and segment overlaps. Table IV summarizes the tracking performance in DER and cpWER evaluated on the seven-channel LibriCSS dataset.

Comparing speaker models, ReDimNet yields the lowest DER (3.38%) and cpWER (5.32%), outperforming the alternatives of ReDimNet2, Wespeaker, NeMo, and SpeechBrain. The performance differences indicate that discriminative embeddings play a constructive role in speaker tracking.

Evaluating VAD models indicates that Pyannote provides the best performance, achieving the lowest DER and cpWER. Silero yields slightly higher DER. Substituting the VAD component by WebRTC significantly increases diarization errors.

Segment length affects speaker tracking. A segment size of 4.0s yields the best DER (3.38%). Enlarging the segment size degrades performance (e.g., 5.35% DER for 8.0s). Longer segments increase the number of speakers within a segment, causing the total number of speakers to potentially surpass the capacity of the speaker separation model ($K = 4$). Conversely, shrinking the segment size to 2.0s also degrades performance (6.75% DER), due to insufficient contextual information for stitching and speaker embedding.

Increasing segment overlap degrades tracking performance. A 25% overlap seems to provide the sufficient context for overlap-based stitching. Larger overlaps compel the system to perform more sequential grouping operations, increasing DER. For example, expanding segment overlap from 25% to 50% raises DER from 3.38% to 4.41%.

Table V provides a detailed breakdown of DER components across several configurations, revealing the impacts of error types on downstream cpWER. The proposed system with different configurations exhibits relatively balanced error distributions, with MI contributions ranging from 1.97% to 2.03%. Compared to the single-channel system, multi-channel system substantially reduces CF errors (from 1.63% to 0.48%). The corresponding results from two SSND versions, one based on TF-GridNet for separation as reported in Table I and another based on SpatialNet for separation as reported in Table II. The second SSND configuration produces higher FA errors (6.98%), but the lower cpWER of 5.13% despite elevated DER (7.62%).

TABLE IV: Ablation study evaluating the impacts of speaker embedding models, VAD, segment lengths and overlaps on DER and cpWER.

$\mathcal{F}_{\text{SPK}}$	$\mathcal{F}_{\text{VAD}}$	L (s)	L_{ovl}/L	DER (%)	cpWER (%)
ReDimNet [44]				3.38	5.32
ReDimNet2 [50]				3.66	5.56
Wespeaker [52]	Pyannote [48]	4.0	0.25	4.18	6.69
NeMo [49]				5.10	8.73
SpeechBrain [51]				5.77	9.33
ReDimNet [44]	Silero WebRTC	4.0	0.25	4.07	5.50
				6.83	5.71
ReDimNet [44]	Pyannote [48]	2.0	0.25	6.75	10.11
		3.0		4.39	6.90
		5.0		3.69	5.72
		6.0		4.14	6.34
		8.0		5.35	7.52
ReDimNet [44]	Pyannote [48]	4.0	0.50	4.41	6.76
			0.75	7.99	11.63

This seemingly contradictory pattern underscores that FA errors during silence periods incur minimal ASR penalty. The oracle configuration achieves the lowest CF (0.48%) and the lowest cpWER (3.66%), highlighting that confusion between speakers remains a bottleneck. These ablation results demonstrate that minimizing MI and CF is essential for speaker tracking systems employed in downstream ASR pipelines, whereas FA errors are secondary.

Compared to SSND [37], our system achieves both high diarization and high ASR results within the same model. In contrast, SSND utilizes separate models for optimal diarization and ASR. Additionally, we employ a sophisticated VAD module to lower FA errors.

TABLE V: Breakdown of DER components (MI, FA, CF) across system configurations and acoustic conditions on LibriCSS dataset.

System	DER	MI	FA	CF	cpWER
SSND (SpatialNet)	7.62	0.53	6.98	0.11	5.13
SSND (TF-GridNet)	4.68	1.86	2.16	0.66	6.84
Speaker tracking					
1-ch, $L = 4$ s, $L_{\text{ovl}}/L = 0.25$	4.63	1.97	1.03	1.63	9.25
7-ch, $L = 5$ s, $L_{\text{ovl}}/L = 0.25$	3.69	2.03	1.04	0.62	5.72
7-ch, $L = 4$ s, $L_{\text{ovl}}/L = 0.25$	3.38	2.02	0.88	0.48	5.32
Oracle, $L = 4$ s, $L_{\text{ovl}}/L = 0.25$	2.98	1.33	1.17	0.48	3.66

Fig. 2 illustrates the speaker count performance of the proposed speaker tracking method on the section-wise LibriCSS dataset. For each section, we run the proposed speaker tracking method and count the number of predicted tracks, which is compared against the actual number of speakers in that section. The results in Fig. 2 are averaged over all sections. The system achieves very high accuracy for mixture audio with few speakers, correctly estimating the speaker count 100% and 97.06% for one and two actual speakers, respectively. As the number of actual speakers increases, the accuracy declines, and the system tends to overestimate the speaker count. For example, with seven actual speakers, the system correctly identifies seven speakers in 62.69% of the segments and estimates eight speakers in 28.36% of segments. This overestimation is primarily attributed to track fragmentation and FA errors, which breaks the same speaker into multiple

tracks. On the other hand, underestimation is infrequent. These results suggest that the proposed framework robustly initiates new speaker tracks, but with a tendency to over-initiate.

Actual number of speakers	8							76.2%	19.0%	4.8%
7				1.5%	4.5%	62.7%	28.4%	3.0%		
6				2.9%	76.3%	18.0%	2.2%	0.7%		
5				1.1%	80.7%	17.1%	1.1%			
4				84.2%	14.1%	1.7%				
3			85.4%	13.5%	1.1%					
2		97.1%	2.9%							
1	100%									
	1	2	3	4	5	6	7	8	9	10
	Estimated number of speakers									

Fig. 2: Speaker counting accuracy of the speaker tracking model evaluated on the section-wise LibriCSS data.

V. CONCLUSIONS

We have proposed a segment-online system for speaker tracking. The system performs speaker separation for simultaneous grouping, yielding the estimates of overlapped speakers. These estimates are further organized sequentially into coherent speakers, each corresponding to a distinct talker. Sequential organization is performed in two stages. The first stage employs overlap-based stitching for continuous organization and the second stage uses working memory based on speaker verification for discontinuous organization. Experimental results show our system advances the state-of-the-art in speaker diarization on LibriCSS. On AMI, we achieve the best diarization results in the segment-online setting. Our system also obtains better speaker-attributed ASR results than other methods, and competitive performance to offline methods. For future work, we plan to explore frame-online processing to enhance real-time processing capability. We also plan to address track fragmentation and false-alarm errors that degrade speaker count accuracy as revealed in Fig. 2.

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